1500–16

Martin Waldseemüller's 1507 world map was the first to name America, although much of the coastline of North and South America remained unknown.

THE EXPLORATION OF NEW LANDS IN THE LATE 15TH CENTURY, and in particular, Columbus's discovery of the Americas in 1492 and Vasco da Gama's circumnavigation of Africa en route to India in 1497–98, provided **much new material for map-makers**.

In 1504, a letter written by Amerigo Vespucci (1454–1512) detailing his third voyage to America came into the hands of a group based in St.-Dié in Lorraine (in modern-day France). One of them, **Martin Waldseemüller** (*c.*1470–1522), produced **a globe and world map in 1507 in which he suggested that the new-found continent be called America**, the first occurrence of the term.

In 1508, Waldseemüller wrote a treatise on surveying, in which **he described the theodolite** (which he called polimetrum) for the first time. Using a theodolite, surveyors and cartographers could now measure angles of up to 360 degrees.

4 YEARS
THE AVERAGE LIFESPAN OF A SWISS MILITARY WATCH BATTERY

40 hours The average lifespan of a 16th-century wind-up pocket watch

Batteries versus springs
Henlein's first watch would have lasted less than two days before needing rewinding, but this was a major achievement at the time.

In the late 15th century, clockmakers learned how to **construct spring-driven clocks** in which the gradual uncoiling of the spring operates the mechanism. The Nuremberg clockmaker **Peter Henlein** (1485–1542) applied this system to portable clocks ("watches"). In 1512, Henlein was recorded as having **made a watch that went for 40 hours and could be carried in a pocket**.

In 1513, Polish astronomer **Nicolaus Copernicus** (1473–1543) wrote his *Commentariolus* (*Little Commentary*), a preliminary outline of **his revolutionary view that Earth revolved in orbit around the Sun**. Feeling dissatisfied with the old planetary theory of Ptolemy, with its multiplicity of celestial spheres, geocentric view, and its anomalies (such as the apparent retrograde motion of some planets), Copernicus explained how a planet's periodicity varied in proportion to its distance from the Sun. Fearing the reaction of the Church, Copernicus kept his findings to himself for 30 years.

Around 1500, **gunsmiths devised the wheel-lock mechanism** for firearms. It used a serrated metal wheel that rotated rapidly, striking against a lump of the mineral pyrite, and creating sparks that lit the gunpowder charge.

First pocket-watch
The compact workings of Peter Henlein's portable clock (c.1512) were driven by a slowly uncoiling spring. It was the first timepiece small enough to be carried in the user's pocket.

1517–35

“FRACASTORO... DECLARED... THAT FOSSIL SHELLS HAD ALL BELONGED TO LIVING ANIMALS.”

Charles Lyell, Scottish geologist,
from *Principles of Geology*, 1830–33

THE MEDIEVAL 13TH-CENTURY SCHOLAR Albertus Magnus had described stones with the "figures of animals," but Arab and medieval scholars believed they were produced by Earth itself or were the remains of animals drowned in the Biblical flood. In a debate in 1517, Italian physician **Girolamo Fracastoro** (1478–1553) was the first to publicly express the view that **fossils are organic matter**, originally animals, that has been ossified over time.

In August 1522, the 18 survivors of Ferdinand Magellan's expedition arrived in Spain, having completed the **first circumnavigation of Earth**. The voyage had taken three years, and more than 230 crew (including Magellan) perished. However, it did definitively prove the size of **Earth's circumference** to be about 24,800 miles (40,000 km).

In 1525, German artist **Albrecht Dürer** (1471–1528) published his *Instructions for Measuring with Compass and Ruler*, **one of the first works on applied mathematics**, which contained detailed accounts of the properties of curves, spirals, and regular and semiregular polygons and solids and their use as **an aid for artists in producing scientifically accurate images**.

Portrait of Paracelsus
Paracelsus was both physician and chemist, and stressed the importance of using chemical techniques in the production of medicines.

German chemist and physician **Theophrastus von Hohenheim** (1493–1541), known as Paracelsus, devised a new classification of chemical substances, rejecting Aristotle and Galen's four humors. In his *De Mineralibus* (*On Minerals*), he divided them instead using the three principal units of sulfur, mercury, and salt. Paracelsus spurned the study of anatomy and promoted the idea that the

1507 Martin Waldseemüller produces first map that uses the name **America**

1508 Waldseemüller describes the **theodolite**

1512 Peter Henlein makes the first **pocket watch**

1513 Nicolaus Copernicus writes his *Commentariolus*, an outline of his **heliocentric theory** of the solar system

1515 The first **wheel-lock pistols** appear in Germany

1517 Frascatoro expresses his view that **fossils** were originally organic life forms

1522 Survivors of Ferdinand Magellan's expedition complete first **circumnavigation** of Earth

1525 Albrecht Dürer publishes first work on **applied mathematics** for use by artists

1525 Christoph Rudolff produces first **German algebraic manual** and introduces modern symbol for square root